

Grade Eight Science
Lab Discussion: What are Variables in an Experiment?

There are three types of variables in science experiments:

- Control Variable(s): Things that are kept the same in all trials of the experiment (all experiments have multiple controls)
- Dependent Variable: What you are measuring in the experiment (the data that you are collecting)
- Independent Variable: What you are changing to get the measurements (the difference between the experimental and control groups)

EXAMPLE:

Research Question: *The Pickle ball vs. The Tennis ball: Which ball will yield the greatest bounce height?*

Background Information	Describe the Background Information
Validity of the Research Question	The purpose of this experiment is to determine which type of ball, a tennis ball or a pickle ball, will bounce the highest when dropped under the same conditions. Tennis and pickleball are both racquet sports that use two racquets, a ball, and a net, but there are some differences, such as the type of ball used. The focus of this investigation is on comparing the bounce height of the tennis ball and the pickle ball to see which one bounces higher and why.
Hypothesis	If a tennis ball and a pickle ball are dropped from a height 100cm then the tennis ball will bounce the highest because tennis balls are made of rubber, while pickle balls are made of plastic, and rubber has more elasticity which will lead to the greatest bounce height
Explanation of the hypothesis	A tennis ball has a rubber core filled with pressurized air (Admin) while a pickle ball is a hollow, plastic ball with holes in it that is not pressurized. (Pickleball Superstore). The experiment will conduct three trials where each type of ball is dropped from a height of 1 meter on the same surface. The data collected will be analyzed, and it will be determined which ball bounces back at a greater height. Research will be conducted to determine why one ball bounces back at a greater height than the other.

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent	The type of ball	The two types of balls considered are a tennis ball and a pickle ball; a new identical ball will be used in each trial
Dependent	The bounce height of the ball	This will be measured using a meter stick attached to the wall and an iPhone camera in slow motion to record the motion
Control	Drop height of the ball (1 m)	During each trial the ball is dropped from a height of 1 m measured using the same meter stick
Control	Surface the ball is dropped on	During each trial the ball is dropped on the same tile floor
Control	Method of recording the ball drop	During each trial the iPhone camera was set to the same setting on slow motion and the same iPhone camera was used

Research Question 1:

Does adding dimples to a car improve the gas mileage?

Background Information	Describe the Background Information
Validity of the Research Question	The purpose of this experiment is to see if dimples on a car cut down on air resistance, therefore, leading to improved gas mileage. Dimples are added to golf balls to enable them to cut through the air in an efficient manner by cutting down on air resistance (Mythbusters). Air resistance is a force exerted on an object slowing it down. ("Air Resistance"). Therefore, this principle is applied to a car to see if dimples will allow the car to move in a more efficient manner, therefore, improving gas mileage by cutting down on air resistance.
Hypothesis	If gas mileage is measured on the same car with dimples and without dimples, then the car with dimples will have better gas mileage than the car without dimples because dimples cut down on air resistance meaning less gas is needed
Explanation of the hypothesis	Air resistance is a force exerted on an object

	slowing it down. ("Air Resistance"). During flight air flow interacts with the surface of the golf ball and can greatly affect the amount of drag. The dimples allow the air to hug the ball more tightly so the air can't detach and pull the ball backward backward. ("Science of Golf: Why Golf Balls Have Dimples"). The hypothesis applies this theory to a car to see if dimples can lead to less air resistance and, therefore, better gas mileage.
--	--

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent	The presence or absence of dimples on the car	Two identical cars were used and each car had a clay mold attached to it; One had a mold of dimples applied to it, while the other did not
Dependent	Gas mileage	The mass of the car was measured before and after driving; the final mass was the mass of the gas burned during the trials of driving
Control(s)	The type of car	The same year, make, and model of car was used, same type and amount of clay applied to each car
Control(s)	The method of measuring the gas mileage	The mass of the car was measured before and after driving; the final mass was the mass of the gas burned during the trials of driving

Research Question 2: ***Are elephants afraid of mice?***

Background Information	Describe the Background Information
Validity of the Research Question	The purpose of this experiment is to see if an elephant is afraid of a mouse in order to prove or disprove this common myth. This type of study may also be valid for scientists studying animal behaviour. The focus of this investigation is to present the elephant with the same situation (a piece of fake elephant dung being knocked over) with and without a mouse present, to see how it reacts to the situation.
Hypothesis	If an elephant is exposed to a piece fake dung being knocked over, with and without a mouse present, then the dung with the mouse under it will cause the elephant to run away in fear because elephants have poor eyesight and may be startled at sudden small movements of the mouse that it is not expecting ("Are Elephants Afraid of Mice?")
Explanation of the hypothesis	Elephants have poor eyesight because they have small eyes that have difficulty focussing images over long distances. They are also color blind to red and green ("Facts About Elephant Eyes"). Therefore, it makes sense to hypothesize that if the elephant cannot see well, that it may not clearly see the mouse, and it may be started by the sudden movement of the mouse.

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent	The presence or absence of the mouse	A piece of fake elephant dung has a mouse placed under it when it is knocked over versus a piece of fake elephant dung that does not have a mouse under it when it is knocked over and the reaction of the elephant is observed
Dependent	The response of the elephant when the mouse is present	This variable is measured using qualitative observations on whether or not the mouse speeds up when moving away from the dung
Control(s)	The set up of the fake elephant dung, mouse and the rock	The dung/mouse set up is placed in the direction of the elephant's food source in each trial; the same piece of fake dung is used and it is set up in the same place in the same

		manner
--	--	--------

MLA Works Cited:

Admin. "The Science Behind Tennis Ball Bounce." *World Tennis*. World Tennis Magazine. 2025. <https://worldtennismagazine.com/the-science-behind-tennis-ball-bounce/24358>

"Air Resistance." *Nikon*. 2022. <https://ngpd.nikon.com/en/glossary/air-resistance.html#:~:text=Air%20resistance%20is%20the%20force,the%20amount%20of%20air%20resistance>. Accessed 09 October, 2025.

"Are Elephants Afraid of Mice?" *The Elephant Sanctuary South Africa*. " <https://www.elephantsanctuary.co.za/blog/140-are-elephants-afraid-of-mice>. Accessed 09 October, 2025.

"Facts About Elephant Eyes." *Siyabona Africa*. " <https://www.krugerpark.co.za/krugerpark-times-4-8-did-you-know-24376.html> Accessed 09 October, 2025

Mythbusters. "Interactive Part 1: Identify the Independent and Dependent Variables with the Mythbusters. *You Tube*. Uploaded by 24/7 Science. 2017. <https://www.youtube.com/watch?v=l0jTMDtX4WY>

Pickleball Superstore. "Are all Pickleballs The Same?" *Pickleball Superstore.com*. Pickleball Superstore. 2024. <https://pickleballsuperstore.com/blogs/pickleball-blog/what-does-a-pickleball-look-like?srltid=AfmBOoqAiOYBj2NixgsMOqPZMI5WO8yWFzSjsOn5phuvEglTRYZOwyvi>

"Science of Golf: Why Golf Balls Have Dimples." *You Tube*. Uploaded by the United States Golf Association. 2015. <https://www.youtube.com/watch?v=fcjaxC-e8oY&t=38s>